

Chapter 121

USD Nodes

This chapter details the Universal Scene Descriptor (USD) nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

Contents

Universal Scene Descriptor (USD) Files	2817	uVisibility [uVis].....	2848
USD Scene Tree Dialog for Object Selection in USD Files	2818	uVolume [uVo].....	2850
uCamera [uCa].....	2819	USD Lights	
uDuplicate [uDp].....	2822	uCylinder Light [uCL].....	2852
uExport [uEx].....	2827	uDisk Light [uDi].....	2853
uImage Plane [uIm].....	2828	uDistant Light [uDL]	2855
uLoader [uLd].....	2830	uDome Light [uDo].....	2856
uMaterialX [uMX].....	2831	uRectangle Light [uRL]	2858
uMerge [uMg].....	2833	uSphere Light [uSL].....	2860
uRenderer [uRn].....	2834	USD Material	
uReplaceMaterial [uRM].....	2837	uNormalMap [uNm].....	2862
uShape [uSh].....	2840	uShader [uSd].....	2863
uSwitch [uSw].....	2843	uTexture [uTx]	2866
uTransform [uXf].....	2845	uTextureTransform [uTXf].....	2867
uVariant [uVa].....	2847	The Common Controls	2868

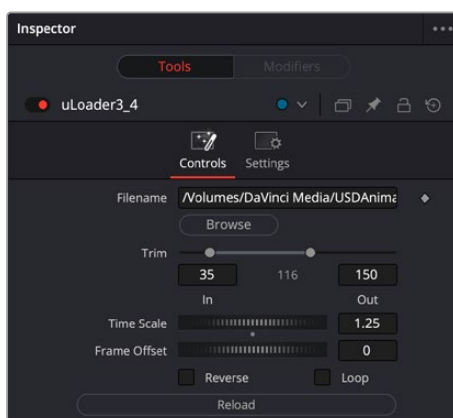
Universal Scene Descriptor (USD) Files

The Universal Scene Descriptor USD framework is a set of open standards for describing, composing, and simulating a 3D environment. The USD file format is more than just a simple format - it's an open source 3D scene description used for creating and sharing 3D content across various applications.

DaVinci Resolve and Fusion can import USD (.usdc, .usdz, and .usda) 3D information including geometry, lighting, cameras, materials, and animation. A new collection of USD tools has been added to Fusion, allowing users to manipulate, re-light, and render these USD files.

The USD Loader

The uLoader allows you to import USD files and adjust its options.



The uLoader node

To Import a USD file:

- 1 Add a uLoader tool to your Fusion Composition.
- 2 Select the USD file you want to load in the File Browser. If you wish to change this file in the future, you can do so in the uLoader Inspector without disrupting your node tree.
- 3 Connect the output of the uLoader to a uMerge tool.
- 4 Connect any other USD tools, like uCamera or any of the uLights, to manipulate the scene.
- 5 Connect the output of the uMerge tool to a uRenderer.
- 6 Connect the output of the uRenderer to the MediaOut tool.

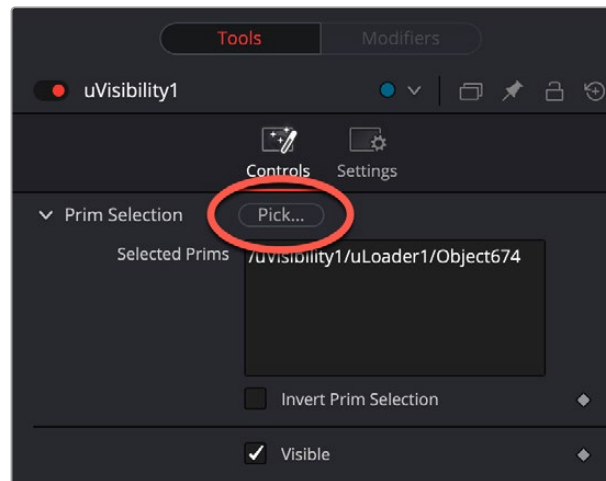


A very simple USD import node tree

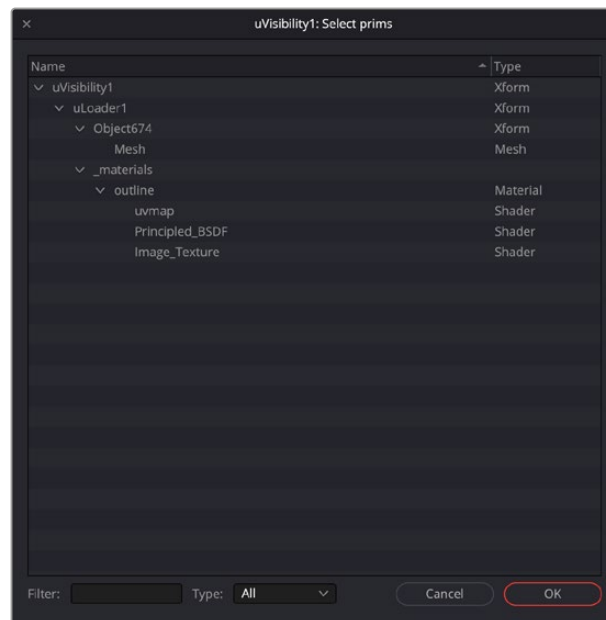
USD Scene Tree Dialog for Object Selection in USD Files

The Scene Tree window is an intuitive visual hierarchy to represent USD scenes. It lets you choose and modify individual elements inside a USD scene. The USD Scene Tree is important for selective effects using the uVisibility, uVariant, uReplace Material, and uTransform nodes, but also used for overriding imported scene objects in uCamera and the various USD light tools.

The USD Scene Tree is activated by pressing the Pick button in the Prim Selection box in certain USD tools.



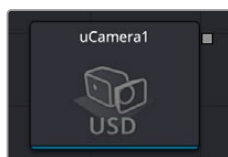
The Prim Selection Pick button



The USD Scene Tree dialog

The tree shows the USD hierarchy, and the type column shows what kind of asset it is. Users can select, filter, isolate, and choose specific elements in a USD scene to focus on logical components and make targeted adjustments. Holding Command allows the toggle selection of multiple objects, and Shift allows selecting a range of objects with a click.

uCamera [uCa]



The USD Camera Node

uCamera Node Introduction

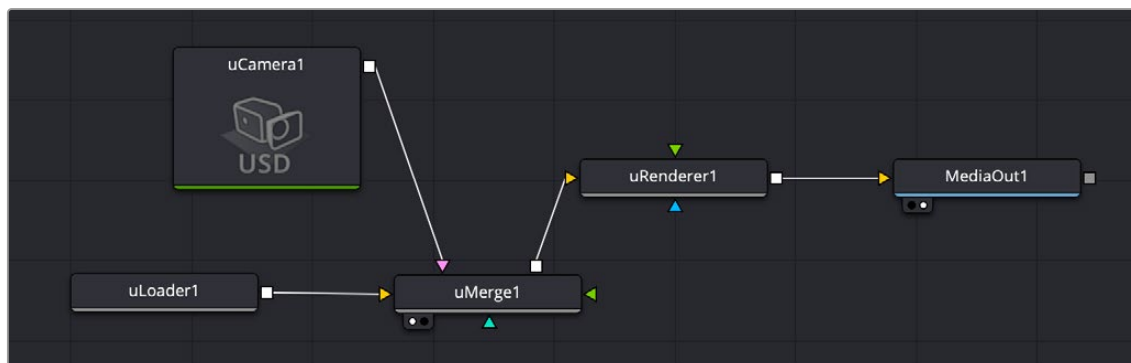
The uCamera node generates a virtual camera for viewing the USD scene in its 3D environment. It closely emulates the settings used in real cameras to make matching live-action or 3D-rendered elements as seamless as possible. Adding any cameras to a 3D composite allows you to frame the elements in a composite how you want and animate the camera during a scene to create moving camera shots.

Inputs

There is one Scene Input in yellow to attach a USD scene. This allows you to adjust any existing cameras within the scene.

Basic Node Setup

The output of a uCamera node should be connected to a uMerge node to build it into a USD 3D scene with other assets and nodes.



The uCamera node is connected to a uMerge node to give standard camera controls to a USD scene.

Displaying a uCamera node directly in the viewer shows only an empty scene; there is nothing for the camera to see. To view the scene through the camera, view the uMerge node where the uCamera is connected, or any node downstream of that uMerge. Then right-click on the viewer and select uCamera > [Camera name] from the contextual menu. Right-clicking on the axis label found in the lower corner of each USD viewer also displays the Camera submenu.

The aspect of the viewer may be different from the aspect of the camera, so the camera view may not match the actual boundaries of the image rendered by the uRenderer node. Guides can be enabled to represent the portion of the view that the camera sees and assist you in framing the shot. Right-click on the viewer and select an option from the Guides > Frame Aspect submenu. The default option uses the format enabled in the Composition > Frame Format preferences. To toggle the guides on or off, select Guides > Show Guides from the viewers' contextual menu, or use the Command-G (macOS) or Ctrl-G (Windows) keyboard shortcut when the viewer is active.

Inspector



The uCamera Controls tab

Controls Tab

The Controls tab contains some of the most fundamental camera settings, including the camera's clipping planes, focal length, and film back properties.

Override Selection

Pressing the Pick button lets you choose an existing camera in an attached USD file to allow adjusting the cameras in an imported scene.

Projection Type

The Projection Type menu is used to select between Perspective and Orthographic cameras. Generally, real-world cameras are perspective cameras. An orthographic camera uses parallel orthographic projection, a technique where the view plane is perpendicular to the viewing direction. This produces a parallel camera output that is undistorted by perspective.

Orthographic cameras present controls only for the near and far clipping planes and a control to set the viewing scale.

Near/Far Clip

The clipping planes are used to limit what geometry in a scene is rendered based on an object's distance from the camera's focal point. Clipping planes ensure objects that are extremely close to the camera, as well as objects that are too far away to be useful, are excluded from the final rendering.

The default perspective camera ignores this setting unless the Adaptive Near/Far Clip checkbox located under the Near/Far Clip control is disabled.

The clip values use units, so a far clipping plane of 20 means that any object more than 20 units from the camera is invisible to the camera. A near clipping plane of 0.1 means that any object closer than 0.1 units is also invisible.

NOTE: A smaller range between the near and far clipping planes allows greater accuracy in all depth calculations. If a scene begins to render strange artifacts on distant objects, try increasing the distance for the Near Clip plane.

Exposure

When selected, the renderer automatically adjusts the camera's Near/Far Clipping plane to match the extents of the scene. This setting overrides the values of the Near and Far clip range controls described above. This option is not available for orthographic cameras.

Focal Length

In the real world, a lens' focal length is the distance from the center of the lens to the film plane. The shorter the focal length, the closer the focal plane is to the back of the lens. The focal length is measured in millimeters. The angle of view and focal length controls are directly related. Smaller focal lengths produce a wider angle of view, so changing one control automatically changes the other to match.

The relationship between focal length and angle of view is $\text{angle} = 2 * \arctan[\text{aperture} / 2 / \text{focal_length}]$. Use the vertical aperture size to get the vertical angle of view and the horizontal aperture size to get the horizontal angle of view.

Focal Distance

Like a focal point on a real-world camera, this setting defines the distance from the camera to an object and is used to calculate depth of field.

F Stop

This is used to define the aperture size of the synthetic lens; it will affect exposure and is used to calculate depth of field.

Film Back

This section allows you to control the technical parameters of the non-lens part of the camera.

Horizontal/Vertical Aperture

The Horizontal Aperture Width and Vertical Aperture Height sliders control the dimensions of the camera's aperture or the portion of the camera that lets light in on a real-world camera. In video and film cameras, the aperture is the size of the opening that defines the area of each frame exposed, also known as sensor size.

Lens Shift X/Y

Provides controls to manually offset the camera lens width and height.

Shutter Close/Open

These define when the shutter of the camera will open and close; this will affect Exposure and Motion Blur.

Stereo Role

If the camera is stereo, this defines the center of role for the two cameras, left, center, or right.

Control Visibility

This section allows you to selectively activate the onscreen controls that are displayed along with the camera.

- **Show View Controls:** Displays or hides all camera onscreen controls in the viewers.
- **Frustum:** Displays the actual viewing cone of the camera.
- **View Vector:** Displays a white line inside the viewing cone, which can be used to determine the center of the frame.
- **Near Clip:** The Near Clipping plane. This plane can be subdivided for better visibility.
- **Far Clip:** The Far Clipping plane. This plane can be subdivided for better visibility.
- **Focal Plane:** The plane set by the Focus Distance. This plane can be subdivided for better visibility.
- **Convergence Distance:** The point of convergence when using Stereo mode. This plane can be subdivided for better visibility.

Common Controls

Transform and Settings Tab

The Transform and Settings tab in the Inspector are also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uDuplicate [uDp]



The USD Duplicate node

Introduction

Similar to the 2D and 3D Duplicate nodes, the uDuplicate node can be used to duplicate any USD asset in a scene, applying a successive transformation to each, and creating repeating models and complex arrays of objects. The options in the Jitter tab allow non-uniform transformations, such as random positioning or sizes.

Inputs

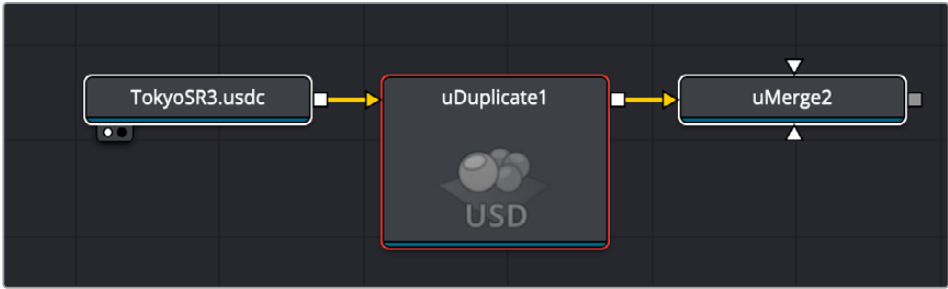
The uDuplicate node has a single input by default where you connect a USD scene. An optional Mesh input appears based on the settings of the node.

SceneInput: The orange Scene Input is a required input. The scene or object you connect to this input is duplicated based on the settings in the Control tab of the Inspector.

MeshInput: A green optional mesh input appears when the Region's tab Region menu is set to mesh. The mesh can be any 3D model, either generated in Fusion or imported.

Basic Node Setup

The output of a uDuplicate node typically connects to a uMerge node, integrating it into a larger scene. The USD scene you want duplicated, in this case a USD scene of a Tokyo shophouse, is connected to the orange input.



A USD scene of a Tokyo shophouse is duplicated.

Usage



The uDuplicate controls

Controls Tab

The Controls tab includes all the parameters you can use to create, offset, and scale copies of the USD object connected to the scene input on the node.

USD Instancing

With this setting enabled, Fusion uses the same USD data for each copy, to increase efficiency. However, in some occasions, when the copies differ too much from each other, it will reduce efficiency instead, so you can disable this setting if that's the case.

Copies

Use this range control to set the number of copies made. Each copy is a copy of the last copy, so if this control is set to [0,3], the parent is copied, then the copy is copied, then the copy of the copy is copied, and so on. This allows some interesting effects when transformations are applied to each copy using the controls below.

Setting the First Copy to a value greater than 0 excludes the original object and shows only the copies.

Time Offset

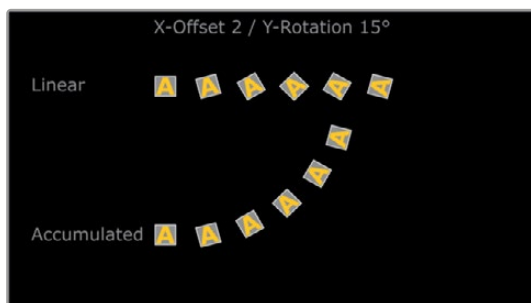
Use the Time Offset slider to offset any animations that are applied to the source geometry by a set amount per copy. For example, set the value to -1.0 and use a cube set to rotate on the Y-axis as the source. The first copy shows the animation from a frame earlier; the second copy shows animation from a frame before that, etc. This can be used with great effect on textured planes—for example, where successive frames of a clip can be shown.

Transform Method



- **Linear:** When set to Linear, transforms are multiplied by the number of the copy, and the total scale, rotation, and translation are applied in turn, independent of the other copies.
- **Accumulated:** When set to Accumulated, each object copy starts at the position of the previous object and is transformed from there. The result is transformed again for the next copy.

Transform Order



With this menu, the order in which the transforms are calculated can be set. It defaults to Scale-Rotation-Transform (SRT).

Using different orders results in different positions of your final objects.

Translation

The X, Y, and Z Offset sliders set the offset position applied to each copy. An X offset of 1 would offset each copy 1 unit along the X-axis from the last copy.

Rotation

The buttons along the top of this group of rotation controls set the order in which rotations are applied to the geometry. Setting the rotation order to XYZ would apply the rotation on the X-axis first, followed by the Y-axis rotation, then the Z-axis rotation.

The three Rotation sliders set the amount of rotation applied to each copy.

Pivot

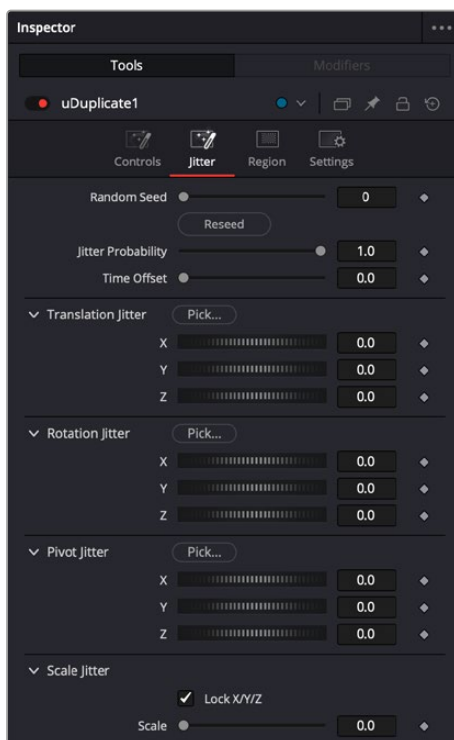
The pivot controls determine the position of the pivot point used when rotating each copy.

Scale

- **Lock:** When the Lock XYZ checkbox is selected, any adjustment to the duplicate scale is applied to all three axes simultaneously. If this checkbox is disabled, the Scale slider is replaced with individual sliders for the X, Y, and Z scales.
- **Scale:** The Scale controls tell Duplicate how much scaling to apply to each copy.

Jitter

The options in the Jitter tab allow you to randomize the position, rotation, and size of all the copies created in the Controls tab.



uDuplicate Jitter tab

Random Seed

The Random Seed slider is used to generate a random starting point for the amount of jitter applied to the duplicated objects. Two Duplicate nodes with identical settings but different random seeds produce two completely different results.

Randomize

Click the Randomize button to auto generate a random seed value.

Jitter Probability

Adjusting this slider determines the percentage of copies that are affected by the jitter. A value of 1.0 means 100% of the copies are affected, while a value of 0.5 means 50% are affected.

Time Offset

Use the Time Offset slider to offset any animations that are applied to the source geometry by a set amount per copy. For example, set the value to -1.0 and use a cube set to rotate on the Y-axis as the source. The first copy shows the animation from a frame earlier; the second copy shows animation from a frame before that, etc. This can be used with great effect on textured planes—for example, where successive frames of a clip can be shown.

Translation Jitter

Use these three controls to adjust the amount of variation in the X, Y, and Z translation of the duplicated objects.

Rotation Jitter

Use these three controls to adjust the amount of variation in the X, Y, and Z rotation of the duplicated objects.

Pivot Jitter

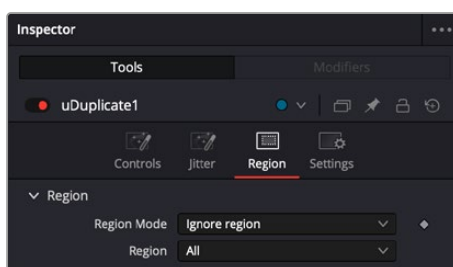
Use these three controls to adjust the amount of variation in the rotational pivot center of the duplicated objects. This affects only the additional jitter rotation, not the rotation produced by the Rotation settings in the Controls tab.

Scale Jitter

Use this control to adjust the amount of variation in the scale of the duplicated objects. Disable the Lock XYZ checkbox to adjust the scale variation independently on all three axes.

Region Tab

The options in the Region tab allow you to define an area in the viewer where the copies can appear or are prevented from appearing. Like most parameters in Fusion, this area can be animated to cause the copied object to pop on and off the screen based on the region's shape and setting.



uDuplicate Region tab

Region Tab

The Region section includes two settings for controlling the shape of the region and the affect the region has on the duplicate objects.

- **Region Mode:** There are three options in the Region Mode menu. The default, labeled “Ignore region” bypasses the node entirely and causes no change to the copies of objects from how they are set in the Controls and Jitter tabs. The menu option labeled “When inside region” causes the copied objects to appear only when their position falls inside the region defined in this tab. The last menu option, “When not Inside region” causes the copied objects to appear only when their position falls outside the region defined in this tab.
- **Region:** The Region menu determines the shape of the region. The five options include cube, sphere, and rectangle primitive shapes. The mesh option allows you to connect a 3D model into the green mesh input on the node. The green input appears only after the Region menu is set to Mesh. The All setting refers to the entire scene. This allows the copies to pop on and off if the Region mode is animated. When the Region menu is set to Mesh, two other options are displayed. These are described below.

Limit by Object ID: When a scene with multiple meshes is connected to the green Mesh input on the node, all the meshes are used as the region. Enabling this checkbox allows you to use the Object ID slider to select the ID for the mesh you want to use as the Region.

Object ID: When the Limit by Object ID checkbox is enabled, this slider selects the number ID for the mesh object you want to use for the Region.

uExport [uEx]



The uExport node

uExport Node Introduction

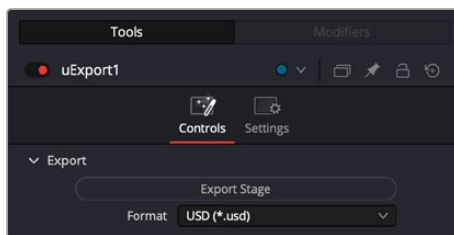
The uExport node is designed to export scenes in the Universal Scene Description (USD) format. The node allows users to export their USD scene, including geometry, materials, animations, and lighting, into a USD file. This upgrade marks an important advancement in Fusion’s functionality, enabling users to seamlessly share complex USD assets, and facilitating smoother workflows in collaborative environments.

The node offers controls to choose between different USD formats such as .usd, .usda, .usdc, and .usdz. It is also possible to export specific nodes in the source tree as a hierarchy of linked USD files.

Inputs

There is one input on the Background node for a USD scene input.

Inspector



The uExport controls

Export

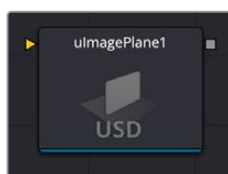
The Export controls let you set the format of the exported USD scene.

- **Export Stage:** Press this button to open a file browser to select where to store your USD scene.
- **Format:** Choose a format for your USD scene.
 - USD (*.usd)
 - USD UTF-8 (*.usda)
 - USD binary (*.usdc)
 - USD packaged (*.usdz)

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

ulmage Plane [ulm]



The USD Image Plane node

ulmage Plane Node Introduction

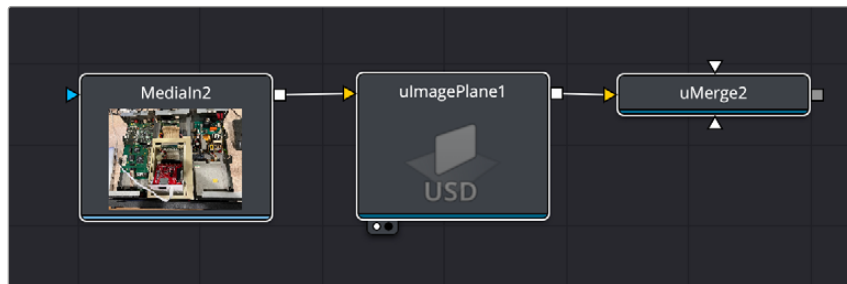
The ulmage Plane node produces 2D planar geometry in 3D space. The node is most commonly used to represent 2D images on “cards” in the 3D space. The aspect of the image plane is determined by the aspect of the image that is input.

Inputs

The input on this node is for the image that will be texture mapped to the plane.

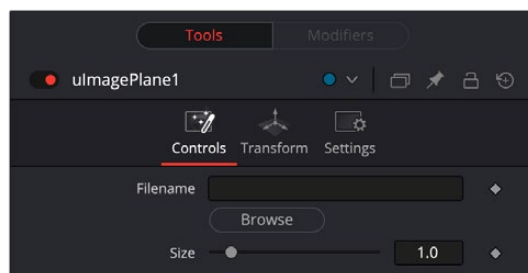
Input: The yellow input is used for the connection of a MediaIn image.

Basic Node Setup



The ulmage Plane Node accepts a 2D image and places it into a 3D USD scene.

Inspector



The ulmage Plane controls

Controls Tab

Most of the Controls tab is taken up by common controls. The Image Plane-specific controls at the top of the Inspector allow minor adjustments.

Filename

The ulmagePlane has an input so an image can be piped directly into the node. This node can also directly load an image by browsing, and it will show the name and path in the filename. Use the Browse button to open the file browser and select an image.

Size

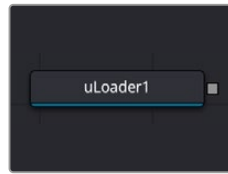
Size will set the size of the image plane in the USD 3D scene.

Common Controls

Transform and Settings Tab

The Transform and Settings tab in the Inspector are also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uLoader [uLd]



The USD Loader node

uLoader Node Introduction

The uLoader node is used for importing USD files into your Fusion Composition. It will load .usd, .usda, .usdc, and .usdz formats.

Replacing Materials with Imported MaterialX files

Users can load MaterialX files (.mtlx) from the the USD Loader and use the uReplaceMaterial tool to assign imported materials to a nominated USD object to achieve a desired look.

Inputs

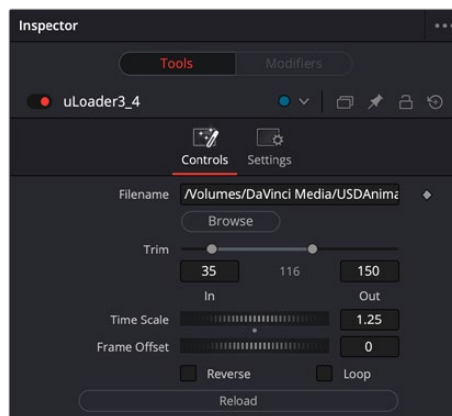
None.

Basic Node Setup



The uLoader node imports a USD scene and then merges it with a USD Camera for view control.

Inspector



The USD Loader controls

Controls Tab

Filename

This control allows you to source a USD file to bring into the Fusion project. Use the Browse button to open the file browser and select a USD file (.usd, .usda, .usdc, or .usdz).

Trim

Lets you select an In and Out point of a USD file to bring only that range into the Fusion project.

Time Scale

Animated .usd scenes can have their animation speed adjusted to be faster and slower using time scale.

Frame Offset

Will define when an animation will start, so timing can be moved along the timeline.

Reverse

When checked, this will reverse the animation of the USD file.

Loop

When checked, this will loop the animation of the USD file.

Reload

This button will reload the USD file set in the Filename field above. This lets you make changes to the USD file in another application and see the results in Fusion by reloading it.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uMaterialX [uMX]



The uMaterialX node

uMaterialX Node Introduction

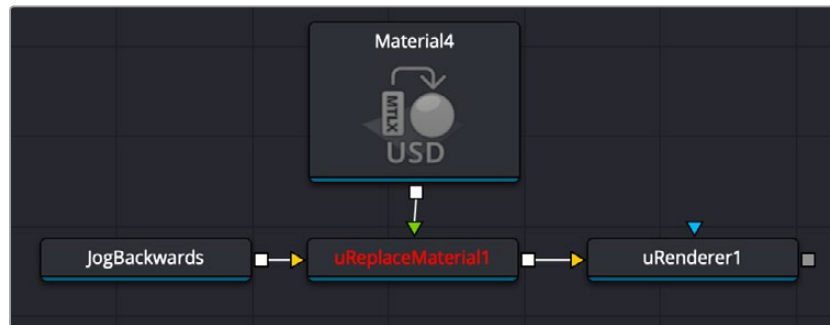
The uMaterialX node simply loads a MaterialX (.mtlx) definition file and creates a USD scene file creating only those materials.

Inputs

There are no inputs to the uMaterialX node. Only a file requester to load the .mtlx file.

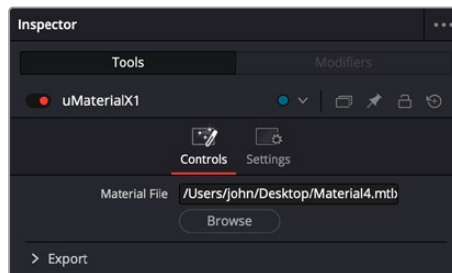
Basic Node Setup

Connect the material output to the rest of the USD tools.



A node tree showing a MaterialX definition (.mtlx) file called Material4 loaded into a uMaterialX node, then passing that material to a uReplaceMaterial node to change the material in a USD scene called JogBackwards.

Inspector



The uMaterialX controls

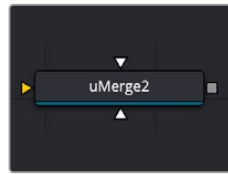
Controls

The single control is a Material File selector with a browse button to load the .mtlx file from your computer.

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uMerge [uMg]



The USD Merge node

uMerge Node Introduction

The uMerge node is the primary node in Fusion that you use to combine separate 3D elements into the same 3D environment.

For example, in a scene created with an image plane, a camera, and a light, the camera would not be able to see the image plane and the light would not affect the image plane until all three objects are introduced into the same environment using the uMerge node.

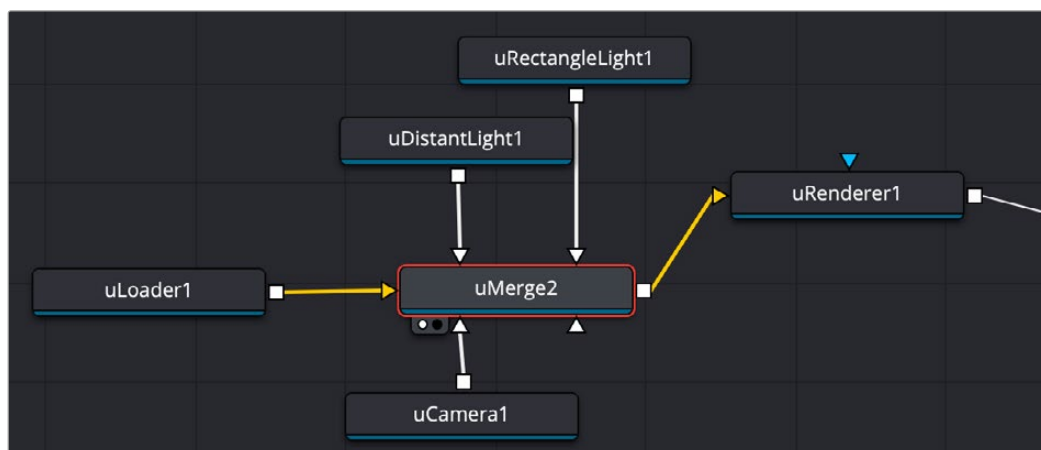
The uMerge provides the standard transformation controls found on most nodes in Fusion's USD suite. Unlike those nodes, changes made to the translation, rotation, or scale of the uMerge affect all the objects connected to the uMerge. This behavior forms the basis for all parenting in Fusion's 3D environment.

Inputs

The uMerge node displays three inputs initially, but as each input is connected a new input appears on the node, assuring there is always one free to add a new element into the scene.

SceneInput[#]: These inputs are used to connect USD image planes, 3D cameras, lights, entire USD scenes, as well as other uMerge nodes. There is no limit to the number of inputs this node can accept. The node dynamically adds more inputs as needed, ensuring that there is always at least one input available for connection.

Basic Node Setup



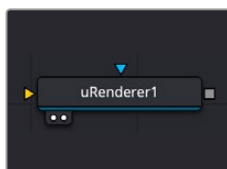
The uMerge node combines the 3D USD scene from the uLoader, two light sources, and the camera, then passes that view out to the uRenderer node to flatten into a 2D image.

Common Controls

Transform and Settings Tab

The Transform and Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uRenderer [uRn]



The uRenderer node

uRenderer Node Introduction

The uRenderer node converts the USD 3D scene into a 2D image, using either a default perspective camera or one of the cameras found in the scene. Every USD scene in a composition terminates with at least one uRenderer node.

Inputs

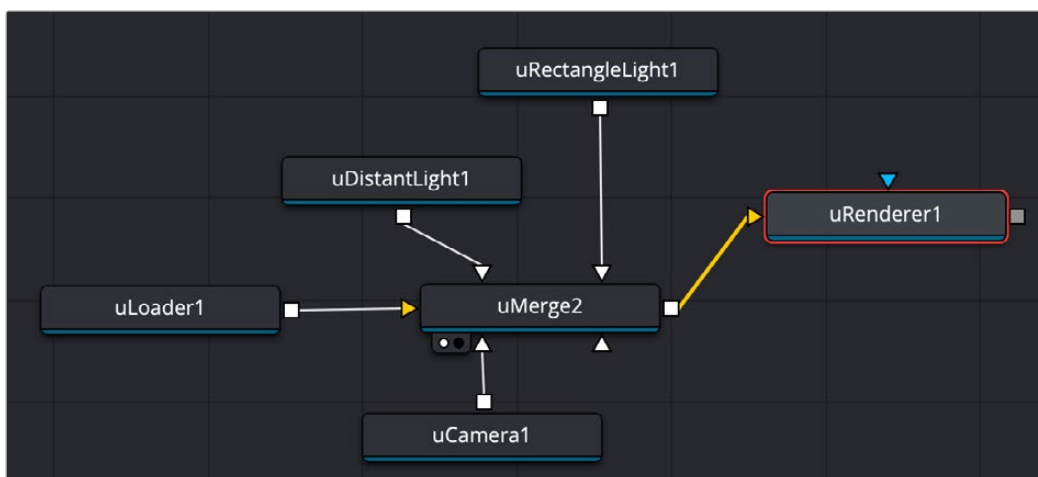
The uRenderer node has two inputs. The main scene input takes in the uMerge or other USD nodes that need to be converted to 2D. The effect mask limits the uRenderer output.

SceneInput: The orange scene input is a required input that accepts a USD scene that you want to convert to 2D.

EffectMask: The blue effects mask input uses a 2D image to mask the output of the node.

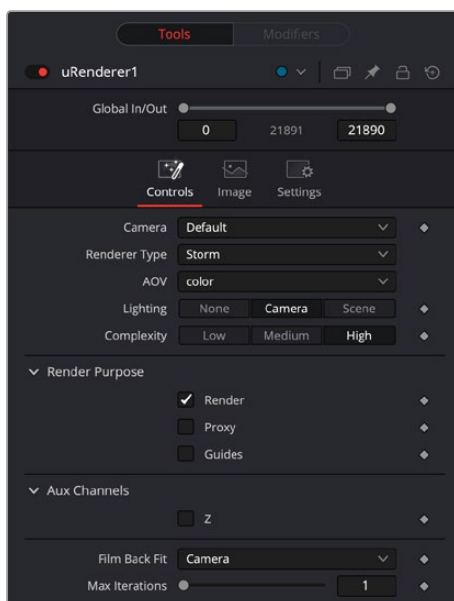
Basic Node Setup

All USD scenes must end with a uRenderer node. The uRenderer node is used to convert a 3D scene into a 2D image. Below, the uRenderer node takes the output of a uMerge node and renders the 3D scene into a 2D image.



The uRenderer node takes the input from the uMerge node and flattens it into a 2D image for display.

Inspector



USD Render controls

Controls Tab

Camera

The Camera menu is used to select which camera from the scene is used when rendering. The Default setting uses the first camera found in the scene. If no camera is located, the default perspective view is used instead.

Renderer Type

This menu lists the available render engines. Storm is currently the only renderer available.

AOV

AOV or Arbitrary Output Variable, are additional passes that the render can output; these are also known as Deep Channels or Auxiliary channels. Color is one; others include Depth and Camera Depth.

- **Color:** This outputs the main full-color render of the USD scene. This will include textures and lighting
- **Depth:** This outputs a black-and-white depth map. This particular depth map is normalized for every frame, showing dynamic black to white values.
- **PrimID:** This outputs a numeric value for each different prim in the USD scene.
- **Camera Depth:** This outputs an accurate camera depth map, which calculates the camera's position relative to the USD scene. The image is rendered as a float 32bit image; at first glance this will appear black.

Lighting

Controls how the USD scene is to be lit.

- **None:** Disables all lighting.
- **Camera:** Enabled objects are lit by the default camera light
- **Scene:** Enabled objects are lit by any lights in the USD scene.
- **Enable Sky Dome:** When checked, the sky dome texture is displayed and rendered from the uRenderer.

Complexity

This defines how much detail is rendered: Low, Medium, or High detail.

Aux Channel Z

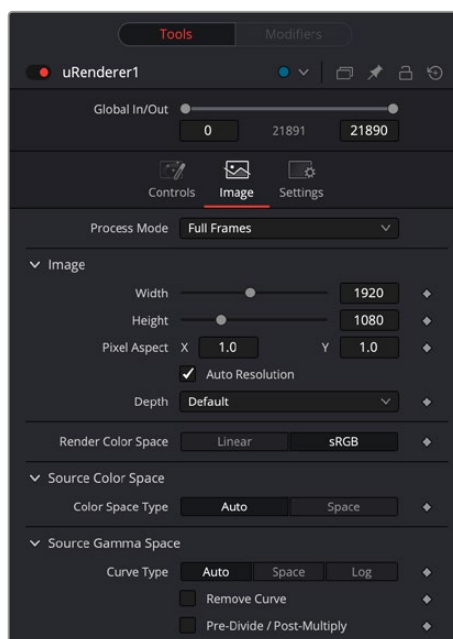
This option enables rendering of the Z channel, the depth channel that represents the distance of each pixel from the camera. When enabled, the camera depth map is rendered to the Z channel.

Film Back Fit

Defines how the rendered image fits the camera's Film Back. If the resolution of the images being rendered is not the same relative dimensions as the film back then these options set how to fit the image.

Max Iterations

This defines the amount of iterations of the rendering algorithm that will be processed.



USD Renderer Image controls

Image Tab

Process Mode: Lets you choose whether the output of the node will be processed as Full Frames or via one of the specified interlaced methods.

Image: Lets you control the output image size and aspect ratio.

- **Width/Height:** Lets you set the pixel dimensions for the output of the node.
- **Pixel Aspect XY:** Lets you set a custom pixel aspect ratio for non-square video formats.
- **Auto Resolution:** Sets the resolution to the Timeline resolution. Uncheck to manually set the width/height above.
- **Depth:** Lets you manually set the bit depth of the output from the uRenderer.

Render Color Space

Allows you to render the output to Linear or sRGB color spaces.

Source Color Space

The Color Space Type menu sets the color space of the media to help achieve a linear workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion but rather adds the source space data into the metadata, if that metadata doesn't already exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver if explicit output spaces are defined there.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows the user to set the color space from a variety of options.

Source Gamma Space

The Curve Type menu automatically determines or allows you to choose the Gamma setting for the image and allows the option to remove the Gamma curve to help achieve a linear workflow.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows you to choose a specific setting from a Gamma Space drop-down menu, while a visual graph lets you see a representation of the gamma setting you've selected.
- **Log:** Similar to the Log-Lin node, this option reveals specific log-encoded gamma profiles so that you can select the one that matches your content. A visual graph shows a representation of the log setting you've selected. When Cineon is selected from the Log Type menu, additional Lock RGB, Level, Soft Clip, Film Stock Gamma, Conversion Gamma, and Conversion table options are presented to finesse the gamma output.
- **Remove Curve:** Depending on the selected gamma space or on the gamma space found in Auto mode, the associated gamma curve is removed from the material, effectively converting it to output in a linear color space.
- **Pre-Divide/Post-Multiply:** Lets you convert "straight" Alpha channels into pre-multiplied Alpha channels, when necessary.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

uReplaceMaterial [uRM]



The uReplaceMaterial node

uReplaceMaterial Node Introduction

The uReplaceMaterial node lets you override an object's material with a new material. This can be a standard color material, texture file, or MaterialX file. The material can be applied to all objects in the scene, or limited to specific objects by using the Prim Selection controls.

Inputs

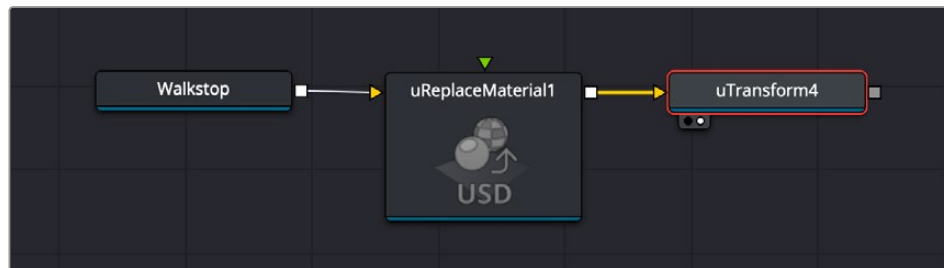
There are two inputs to the uReplaceMaterial node.

Scene Input: The yellow input accepts a USD scene file from a uLoader tool.

Material Input: The green input accepts material files from various tools like uMaterialX, uShader, or even picked from another USD scene using a uLoader tool.

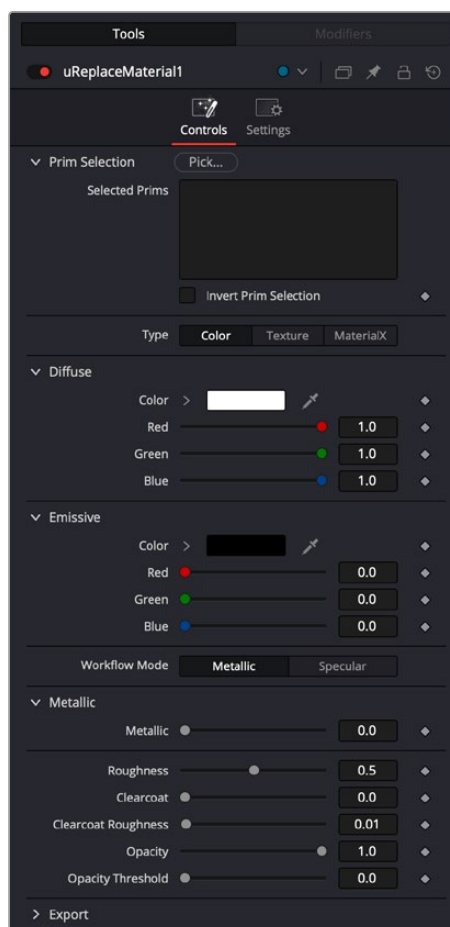
Basic Node Setup

Connect the material output to the rest of the USD tools.



A simple node tree showing a uReplaceMaterial tool accepting input from a USD scene called Walkstop, replacing a material in the scene, then passing that scene to a uTransform node to adjust its transform parameters.

Inspector



The uReplaceMaterial controls

Controls

The uReplaceMaterial controls let you change the materials used in a USD scene.

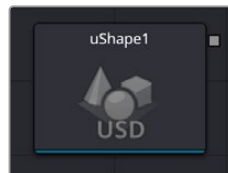
- **Prim Selection:** Pick opens the Scene Tree window that lets you choose specific materials or objects in a USD scene.
 - Invert Prim Selection:** Checking this box inverts the Prim Selection to exclude the materials or objects in the Selected Prims box.
- **Type:** Select the type of material you want to replace the original with.
 - Color:** Chooses a standard Color material and opens the color controls.
 - Texture:** Presents a file requester for loading a texture file.
 - MaterialX:** Presents a file requester for loading a MaterialX file, or choosing a Material from a file attached to the green Material input on the node. Choosing this option closes all other controls.
- **Diffuse:** Lets you change the color of the material, or load a texture.
 - Color:** Opens a standard color picker to choose a new color for the material.
 - Red:** Lets you adjust the amount of red in the Material.
 - Green:** Lets you adjust the amount of green in the Material.
 - Blue:** Lets you adjust the amount of blue in the Material.
- **Emissive:** Emissive adds a global lighting effect to a shape, creating a layer of color over the existing texture. It can be adjusted to different levels of intensity and can be used to simulate the emission of light.
 - Color:** Opens a standard color picker to choose a new color for the lighting.
 - Red:** Lets you adjust the amount of red in the lighting.
 - Green:** Lets you adjust the amount of green in the lighting.
 - Blue:** Lets you adjust the amount of blue in the lighting.
- **Workflow Mode:** Lets you set the mode between Metallic and Specular.
- **Metallic:** When enabled, this control shows the Metallic slider. The Metallic slider adds a reflective quality to objects, creating the appearance of metal. It enhances the reflective properties of a shape.
- **Specular:** When enabled, this shows the Specular Color controls. Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears.
- **Roughness:** Affects the amount of light reflection and scattering on a surface, giving it a smoother or more textured appearance, depending on the setting. It is a key element in creating realistic reflections.
- **Clearcoat:** Affects the appearance of a surface by adding a glossy layer on top of it, mimicking the effect of a protective coating. It is commonly used in creating materials like car paint or polished metal.
- **Clearcoat Roughness:** Adjusts the level of imperfection or smoothness of the glossy layer added on top of a surface. This can affect the realism and reflection of materials like car paint or polished metal.
- **Opacity:** Reduces the material and object opacity, impacting the color and Alpha values of the shape.
- **Opacity Threshold:** Adjusts where the opacity starts to appear.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uShape [uSh]



The USD Shape node

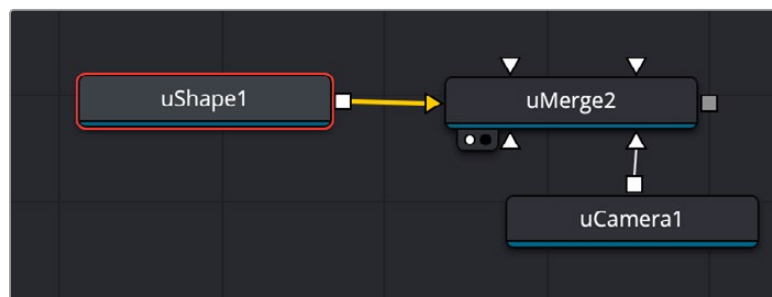
uShape Node Introduction

The uShape node is used to produce several basic primitive 3D shapes, including capsule, torus, planes, cubes, spheres, and cylinders.

Inputs

None.

Basic Node Setup



The uShape node generates a 3D shape that is merged with a uCamera node to adjust the view.

Inspector



The USD Shape Controls

Controls Tab

Shape

This menu allows you to select the primitive geometry produced by the Shape 3D node. The remaining controls in the Inspector change to match the selected shape.

The Shape primitives are Capsule, Cone, Cube, Cylinder, Ico sphere, Plane, Sphere and Torus.

Double Sided

This will make the polygons have two faces and receive lighting on both sides.

Lock Width/Height/Depth

Only for Plane and Cube shapes. If this checkbox is selected, the width, height, and depth controls are locked together as a single size slider. Otherwise, individual controls over the size of the shape along each axis are provided.

Size Width/Height/Depth

Only for Plane and Cube shapes. Used to control the size of the shape.

Radius

When a Capsule, Sphere, Cylinder, Cone, or Torus is selected in the shape menu, this control sets the radius of the selected shape.

Height

When a Capsule, Cone, or Cylinder is selected in the shape menu, this control sets the height of the selected shape.

Top Radius

When a cone is selected in the Shape menu, this control is used to define a radius for the top of a cone, making it possible to create truncated cones.

Subdivision Level/Base/Height/Cap/Cylinder

The Subdivision controls are used to determine the tessellation of the mesh on all shapes. The higher the subdivision, the more vertices and polygons in each shape.

Subdivision Scheme

The Subdivision Scheme defines which algorithm is used to tessellate the polygons in the shapes. The methods are None, Bilinear, Loop and Catmull-Clark.

Angle

When the Capsule, Cone, Cylinder, Sphere, or Torus shape is selected in the Shape menu, this range control determines how much of the shape is drawn. A start angle of 180° and end angle of 360° would only draw half of the shape.

Latitude

When a Sphere or Torus is selected in the Shape menu, this range control is used to crop or slice the object by defining a latitudinal subsection of the object.

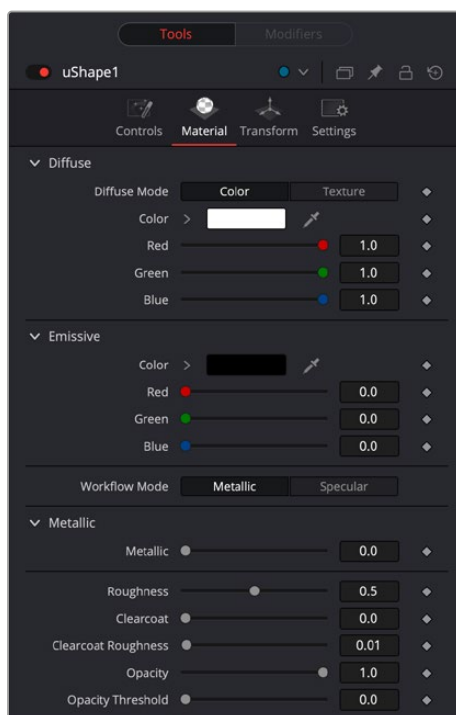
Cap Bottom/Top

When Cylinder or Cone is selected in the Shape menu, the Bottom Cap and Top Cap checkboxes are used to determine if the end caps of these shapes are created or if the shape is left open

Section

When the Torus is selected in the Shape menu, Section controls the thickness of the tube making up the torus.

Material Tab



The USD Shape Material controls

Diffuse Mode

Diffuse describes the base surface characteristics without any additional effects like reflections or specular highlights.

- **Color:** Provides controls to modify the shape's diffuse color.
- **Texture:** This control allows you to use a source image as the diffuse surface color. Use the Browse button to open the file browser and select an image file.

Emissive

Emissive adds a global lighting effect to a shape, creating a layer of color over the existing texture. It can be adjusted to different levels of intensity and can be used to simulate the emission of light.

Workflow Mode

Lets you set the mode between Metallic and Specular.

- **Metallic:** When enabled, this control shows the Metallic slider. The Metallic slider adds a reflective quality to objects, creating the appearance of metal. It enhances the reflective properties of a shape.
- **Specular:** When enabled, this shows the Specular Color controls. Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears.

Roughness

Affects the amount of light reflection and scattering on a surface, giving it a smoother or more textured appearance depending on the setting. It is a key element in creating realistic reflections.

Clearcoat

Affects the appearance of a surface by adding a glossy layer on top of it, mimicking the effect of a protective coating. It is commonly used in creating materials like car paint or polished metal.

Clearcoat Roughness

Adjusts the level of imperfection or smoothness of the glossy layer added on top of a surface. This can affect the realism and reflection of materials like car paint or polished metal.

Opacity

Reduces the material and object opacity, impacting the color and Alpha values of the shape.

Common Controls

Transform and Settings Tab

The Transform and Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uSwitch [uSw]



The uSwitch node

uSwitch Node Introduction

uSwitch functions similarly to Switch but is specifically designed for the USD environment.

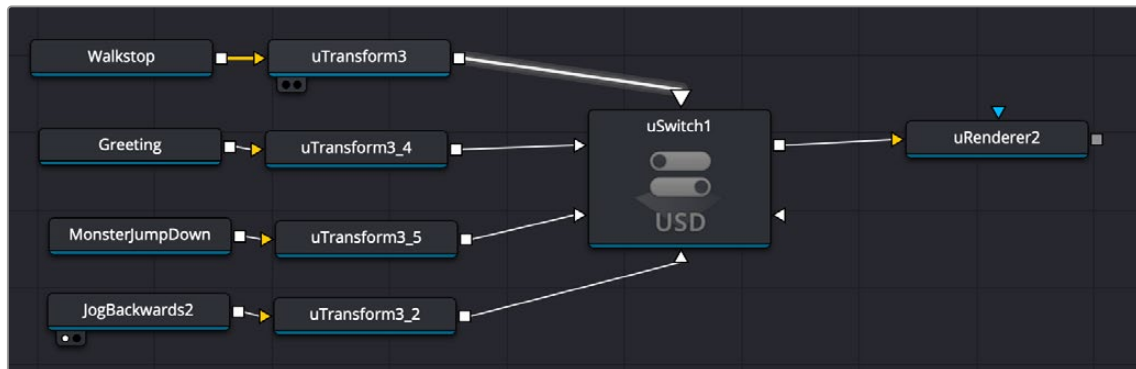
uSwitch is a tool that enables artists to alternate between multiple USD input sources, allowing the selection of one output from the chosen inputs. This capability is especially useful for toggling between different visual elements within a composition. The number of input connections can vary, with the node supporting a dynamic number of inputs that can be renamed as needed. Controls in the Config tab allow users to add or remove inputs and change their names.

Inputs

You can add up to nine inputs to a uSwitch node, using the slider in the Config tools. If you need more than nine sources, you can type that number manually into the Number of Inputs field.

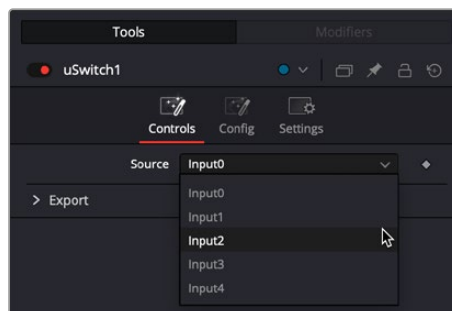
Basic Node Setup

Connect as many input sources as you wish to the uSwitch Node and connect its output to the rest of the node tree.



The uSwitch tool receiving five USD media inputs, allowing you to choose one to go out to uRender.

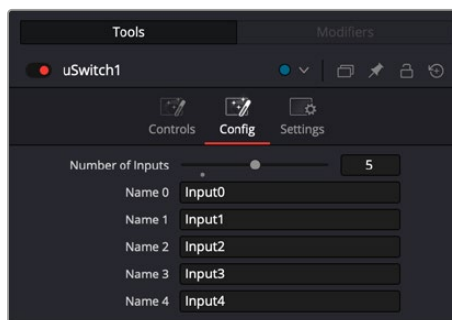
Inspector



The uSwitch controls let you choose a source to pass through the switch.

Controls

The Switch Controls consist of a Source switcher that lets you choose which source to pass through the switch.



The uSwitch configuration

Config

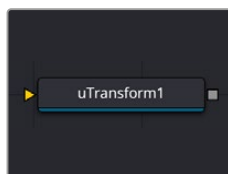
The uSwitch configuration lets you choose the number of inputs and rename them.

- **Number of Inputs:** Adjust the slider to add the number of inputs you wish to use, up to nine. If you want more than nine inputs, you can click in the number field and type the number manually.
- **Name X:** You can change the name of the input from numerical to something more descriptive.

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uTransform [uXf]



The USD Transform node

uTransform Node Introduction

If you have multiple objects or prims in your USD scene, the uTransform node allows you to choose individual groups of objects and adjust their transform independently from the rest of the scene.

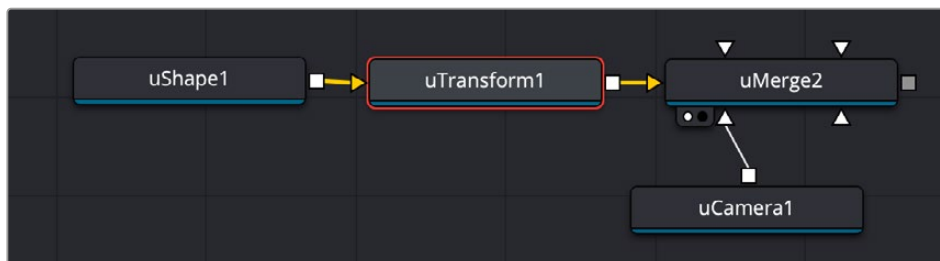
Inputs

The uTransform node has a single required input for a USD scene or USD object.

Scene Input: The orange scene input is connected to a USD scene or USD object to apply a second set of transformation controls.

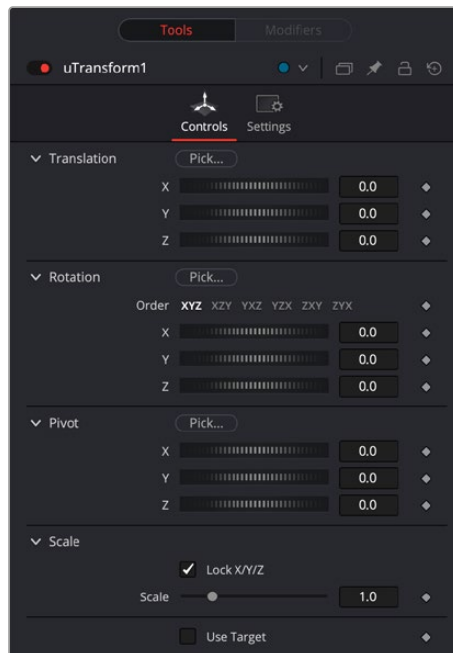
Basic Node Setup

The uTransform node adds 3D position, rotation, and pivot control onto any existing transforms in the USD node prior to it. You can combine multiple uTransform nodes together to build parenting or hierarchical movement.



The uTransform node gives you another set of 3D transform controls that can be used to modify those in the Transform tab on the uShape node.

Inspector



The USD Transform controls

Controls Tab

Pick

Pick opens the Scene Tree window that lets you choose specific materials or objects in a USD scene. For more information on using the *USD Scene Tree*, see its section at the beginning of this chapter.

Translation

Transform controls are used to position the USD shape in 3D space.

Rotation Order

Use these buttons to select the order used to apply the rotation along each axis of the object. For example, XYZ would apply the rotation to the X-axis first, followed by the Y-axis, and then the Z-axis.

Rotation

Use these controls to rotate the shape around its pivot point. If the Use Target checkbox is selected, then the rotation is relative to the position of the target; otherwise, the global axis is used.

Pivot

A pivot point is the point around which an object rotates. Normally, an object rotates around its own center, which is considered to be a pivot of 0,0,0. These controls can be used to offset the pivot from the center.

Scale

If the Lock X/Y/Z checkbox is checked, a single scale slider is shown. This adjusts the overall size of the object. If the Lock checkbox is unchecked, individual X, Y, and Z sliders are displayed to allow scaling in any dimension.

Scale

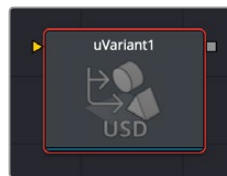
Selecting the Use Target checkbox enables a set of controls for positioning an XYZ target. When the target is enabled, the object always rotates to face the target. The rotation of the object becomes relative to the target.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other USD nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

uVariant [uVa]



The uVariant node

uVariant Node Introduction

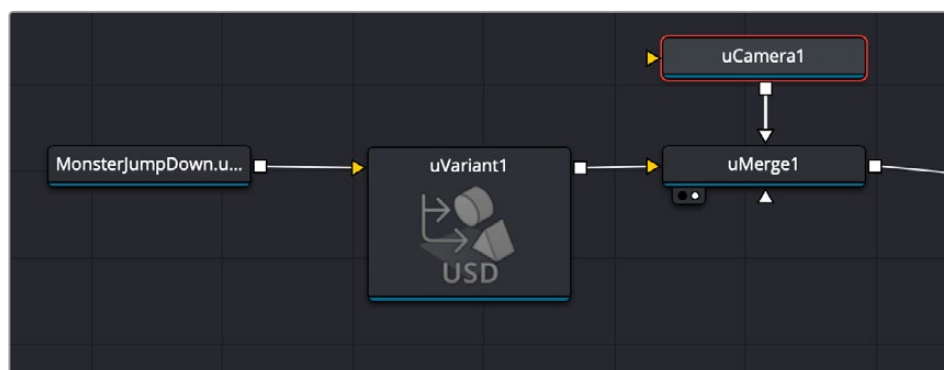
Connecting the uVariant node to the USD loader lets you switch between different alternatives or “variants” for assets that have been set in the USD scene.

Inputs

The uVariant node has a single required input for a USD scene or USD object.

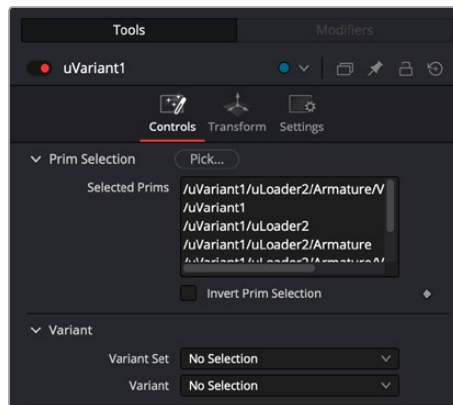
Basic Node Setup

The uVariant node lets you choose between any of the set variants created inside a USD scene. It accepts a USD scene input.



The uVariant node accepts the input from the MonsterJumpDown USD scene.

Inspector



The uVariant controls

Controls Tab

Pick

Pick opens the Scene Tree window that lets you choose specific materials or objects in a USD scene. For more information on using the *USD Scene Tree*, see its section at the beginning of this chapter.

Selected Prims

Shows a list of the prims that have been picked.

Invert Prim Selection

Check this box to swap the selected prims with the unselected ones. This action is animatable using keyframes.

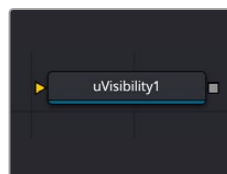
Variant Set

This drop-down menu will show which of the scene elements have variants to choose from. If no variants are available it will show No Selection.

Variant

Once you have selected a variant set above, the variant options to choose between will appear in this dropdown.

uVisibility [uVis]



The uVisibility node

uVisibility Node Introduction

Connecting the uVisibility node to the USD loader lets you control the visibility of an object or branch of objects in a scene.

Inputs

The uVisibility node has a single required input for a USD scene or USD object.

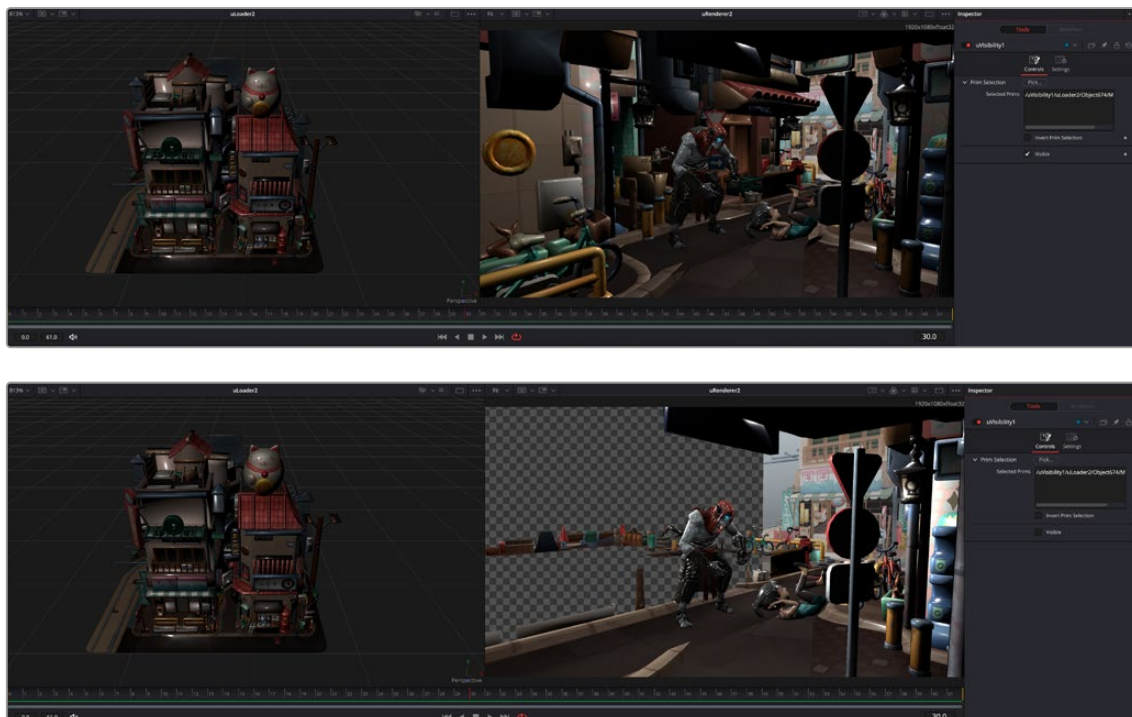
Scene Input: The orange scene input is connected to a USD scene or USD object to apply a second set of transformation controls.

Basic Node Setup

The uVisibility node accepts a USD scene and then outputs the visible elements to the rest of the node tree.

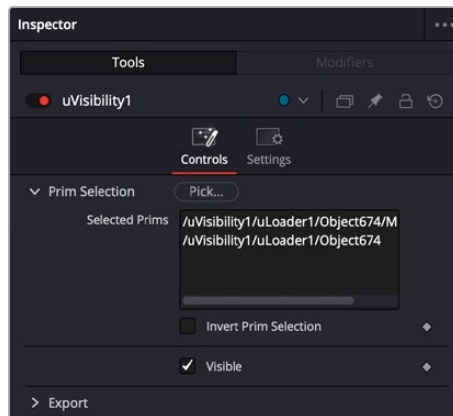


The uVisibility node



Using the uVisibility tool, you can show or hide individual 3D assets. In the above, the cityscape model (left) has been marked visible, and below it is hidden with the visible box unchecked.

Inspector



The uVisibility controls

Controls Tab

Pick

Pick opens the Scene Tree Dialog that lets you choose specific materials or objects in a USD scene. As you select the prims, they will highlight in the viewer. See the beginning of this chapter for information on using the Scene Tree Dialog.

Selected Prims

Shows a list of the prims that have been picked.

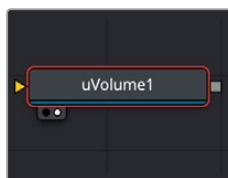
Invert Prim Selection

Check this box to swap the selected prims with the unselected ones. This action is animatable using keyframes.

Visible

Toggling this checkbox will add or remove the selected prims from the view. This action is animatable using keyframes.

uVolume [uVo]



The uVolume tool

Introduction

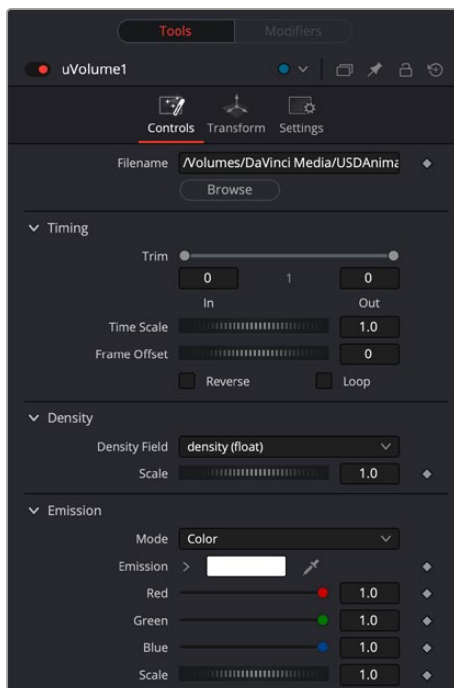
uVolume is a versatile and powerful tool that simplifies the process of importing volumetric VDB files into a USD environment. This node allows you to adjust VDB parameters such as density, temperature, and color, giving you complete control over the appearance and behavior of the imported VDB file.

Usage

When uVolume is added to the Node Editor, a File dialog is displayed automatically allowing the selection of a VDB file from your media drive.

You can add either a single VDB file or a VDB sequence (animated VDB).

Once a VDB is brought in using the uVolume node, the uVolume can be used for trimming, looping, and changing the speed of the imported asset.



The uVolume controls for manipulating VDB files

You can adjust the density with the Scale control, allowing for precise and fine-tuned adjustments.

The Emission section provides color controls for the VDB. Using the Mode selection pulldown will give you the option to select a method for how to change the color of the VDB. Emission modes include:

- **Color:** Applies an overall 'diffuse' color to import VDB. Note, USD lights will not affect a VDB with color set to white.
- **Field:** Scatters an overall color to a nominated Emission Field.
- **Blackbody:** Applies color and intensity based on temperature; this is suitable for rendering fire and explosions.
- **Gradient:** Allows you to map a range of colors over a specified field. The Gradient option provides you with a number of controls to target a field, then apply nominated colors to that field.

The uVolume provides the standard transformation controls found on most nodes in Fusion's USD suite.

USD Lights

USD light tools are a powerful addition to Fusion's USD toolset. They allow for greater control and flexibility when creating and manipulating lighting in your USD scenes. There are six lights available in the USD toolset, they are:

uCylinder Light [uCL]



The USD Cylinder light

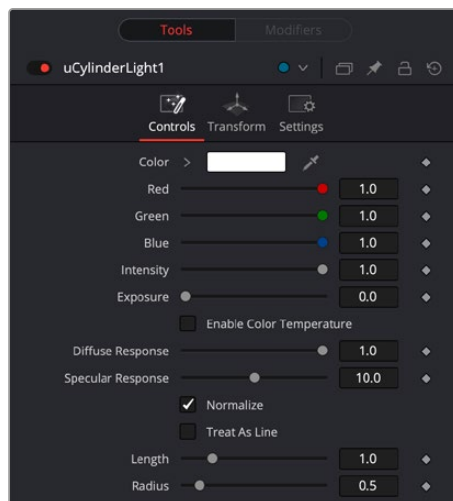
uCylinder Light Node Introduction

A uCylinder light is geometric light that has length and diameter, similar to fluorescent light. This light shows an onscreen control. Translation and rotation of the control place this in the scene.

Inputs

There is one Scene Input in yellow to attach a USD scene; this allows you to adjust any existing lights within the scene.

Inspector



The USD Cylinder Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uCylinder light. The position, direction, and scale of the light source is controlled in the Transform tab.

Override Selection

Pressing the Pick button lets you choose an existing light in an attached USD file to allow adjusting the lights in an imported scene.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20% percent light.

Exposure

This will change how much light will expose a scene; this is similar to Intensity.

Color Temperature/Enable

This sets the color temperature of a light source. Its default is 6500K, which is daylight temperature.

Diffuse Response

Controls the amount the light will contribute to the Diffuse color of a material.

Specular Response

Controls the amount the light will contribute to the Specular color of a material.

Normalize

This will normalize the light's contribution in the scene.

Treat As Line

This will make the light a simpler line light source.

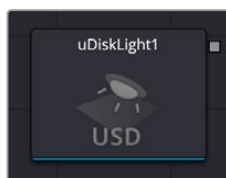
Length

Defines the length of the light.

Radius

Defines the diameter of the light.

uDisk Light [uDi]



The USD Disk light

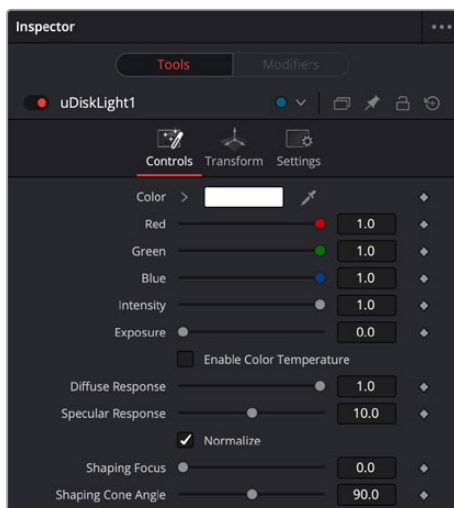
uDisk Light Node Introduction

A uDisk light is a round flat light, similar to an umbrella or softlight. The Controls tab is used to set the color and brightness of the uDisk light. The position, direction, and scale of the light source is controlled in the Transform tab.

Inputs

There is one Scene Input in yellow to attach a USD scene; this allows you to adjust any existing lights within the scene.

Inspector



The USD Disk Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uDisk light. The position, direction, and scale of the light source is controlled in the Transform tab.

Override Selection

Pressing the Pick button lets you choose an existing light in an attached USD file to allow adjusting the lights in an imported scene.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20% percent light.

Exposure

This will change how much light will expose a scene; this is similar to Intensity.

Color Temperature / Enable

This sets the color temperature of a light source. Its default is 6500K, which is daylight temperature.

Diffuse Response

Controls the amount the light will contribute to the Diffuse color of a material.

Specular Response

Controls the amount the light will contribute to the Specular color of a material.

Normalize

This will normalize the light's contribution in the scene.

Shaping Focus

This defines focus point of the lens effect of this light.

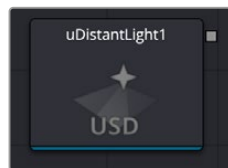
Shaping Cone Angle

Defines the spread of the light.

Radius

Defines the diameter of the light.

uDistant Light [uDL]



The USD Distant light

uDistant Light Node Introduction

A uDistant light is similar to a directional light and simulates distance source, similar to sunlight. This light shows an onscreen control. The rotation of the control is used to determine from where in the scene the light direction is pointing.

Inputs

There is one Scene Input in yellow to attach a USD scene; this allows you to adjust any existing lights within the scene.

Inspector



The USD Distant Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uDistant light. The position, direction, and scale of the light source is controlled in the Transform tab.

Override Selection

Pressing the Pick button lets you choose an existing light in an attached USD file to allow adjusting the lights in an imported scene.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20% percent light.

Exposure

This will change how much light will expose a scene; this is similar to Intensity.

Color Temperature/Enable

This sets the color temperature of a light source. Its default is 6500K, which is daylight temperature.

Diffuse Response

Controls the amount the light will contribute to the Diffuse color of a material.

Specular Response

Controls the amount the light will contribute to the Specular color of a material.

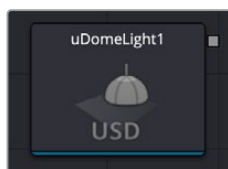
Normalize

This will normalize the light's contribution in the scene.

Angular Size

Defines the spread of the light.

uDome Light [uDo]



The USD Dome light

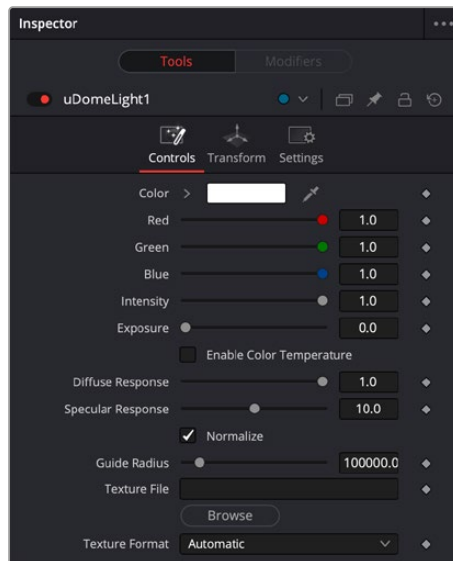
uDome Light Node Introduction

A uDome light uses SDR and HDR images and surrounds the entire scene in a sphere light mapped with the image, similar to environment light. This light shows an onscreen control. The rotation of the control is used to determine from where in the scene the light direction is pointing.

Inputs

There is one Scene Input in yellow to attach a USD scene; this allows you to adjust any existing lights within the scene.

Inspector



The USD Dome Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uDome light. The position, direction, and scale of the light source is controlled in the Transform tab.

Override Selection

Pressing the Pick button lets you choose an existing light in an attached USD file to allow adjusting the lights in an imported scene.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20% percent light.

Exposure

This will change how much light will expose a scene; this is similar to Intensity.

Color Temperature/Enable

This sets the color temperature of a light source. Its default is 6500K, which is daylight temperature.

Diffuse Response

Controls the amount the light will contribute to the Diffuse color of a material.

Specular Response

Controls the amount the light will contribute to the Specular color of a material.

Normalize

This will normalize the light's contribution in the scene.

Guide Radius

This sets how large the dome is around the scene. The default is set to large and far away, similar to the outdoors. This radius can be set smaller to simulate a room size environment.

Texture File/Browse

This will load an image to map on the Domelight.

Texture Format

Sphere mapped images can be stored in different schemes; the common schemes are Lat-Long, MirrorBall, Angular, and Cube Mapped Vertical Cross.

uRectangle Light [uRL]



The USD Rectangle light

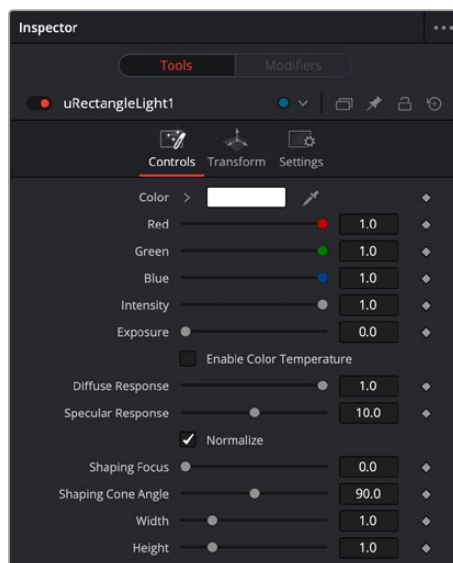
uRectangle Light Node Introduction

A uRectangle light is a flat light, similar to softlight or window . The Controls tab are used to set the color and brightness of the uRectangle light. The position, direction, and scale of the light source is controlled in the Transform tab.

Inputs

There is one Scene Input in yellow to attach a USD scene; this allows you to adjust any existing lights within the scene.

Inspector



The USD Rectangle Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uDome light. The position, direction, and scale of the light source is controlled in the Transform tab.

Override Selection

Pressing the Pick button lets you choose an existing light in an attached USD file to allow adjusting the lights in an imported scene.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20% percent light.

Exposure

This will change how much light will expose a scene; this is similar to Intensity.

Color Temperature/Enable

This sets the color temperature of a light source. Its default is 6500K, which is daylight temperature.

Diffuse Response

Controls the amount the light will contribute to the Diffuse color of a material.

Specular Response

Controls the amount the light will contribute to the Specular color of a material.

Normalize

This will normalize the light's contribution in the scene.

Shaping Focus

This defines focus point of the lens effect in this light.

Shaping Cone Angle

Defines the angle of spread of the light.

Width

Defines how wide the light appears.

Height

Defines how high the light appears.

uSphere Light [uSL]



The USD Sphere light

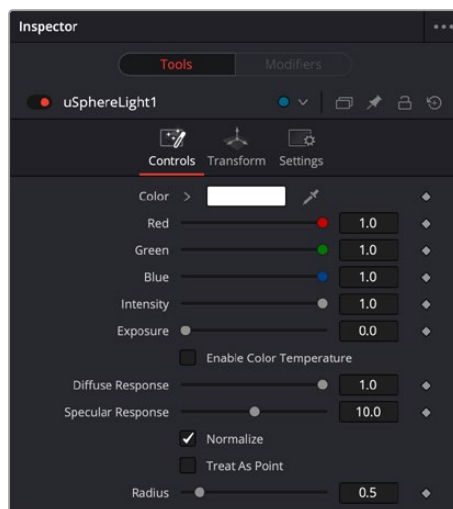
uSphere Light Node Introduction

A uSphere light is local light that is similar to point light but also has radius. This light shows an onscreen control. The rotation of the control is used to determine from where in the scene the light direction is pointing.

Inputs

There is one Scene Input in yellow to attach a USD scene; this allows you to adjust any existing lights within the scene.

Inspector



The USD Sphere Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uSphere light. The position, direction, and scale of the light source is controlled in the Transform tab.

Override Selection

Pressing the Pick button lets you choose an existing light in an attached USD file to allow adjusting the lights in an imported scene.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20% percent light.

Exposure

This will change how much light will expose a scene; this is similar to Intensity.

Color Temperature/Enable

This sets the color temperature of a light source. Its default is 6500K, which is daylight temperature.

Diffuse Response

Controls the amount the light will contribute to the Diffuse color of a material.

Specular Response

Controls the amount the light will contribute to the Specular color of a material.

Normalize

This will normalize the light's contribution in the scene.

Treat as Point

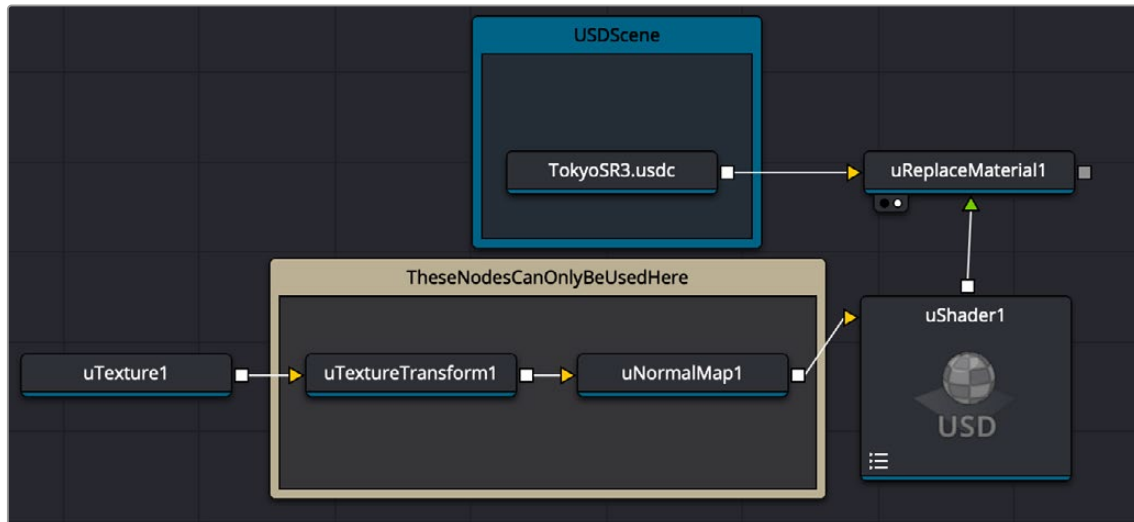
This will make the light a simple classic point source.

Radius

This sets the size of the sphere light.

USD Material

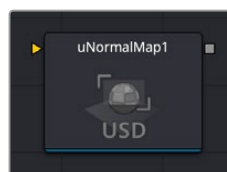
The `uTexture`, `uTextureTransform`, `uNormalMap`, and `uShader` nodes give you the ability to import and manipulate textures directly in your USD scene without having to wrap them in a USD file first. These tools add flexibility and control to your USD workflow within the Fusion page.



A sample texture replacement node tree using the USD Materials tools

The `uTexture`, `uTextureFormat`, and `uNormalMap` tools are designed to work around and feed a texture to the `uShader` tool, where it is manipulated using either the USD or MaterialX models to change its appearance. The `uShader` is commonly connected to a `uReplaceMaterial` node, which selects the specific parts of the entire USD scene to apply the new texture to. This workflow lets you quickly iterate texture changes internally in Fusion, without requiring the modification of the original USD scene in external 3D modeling software.

uNormalMap [uNm]

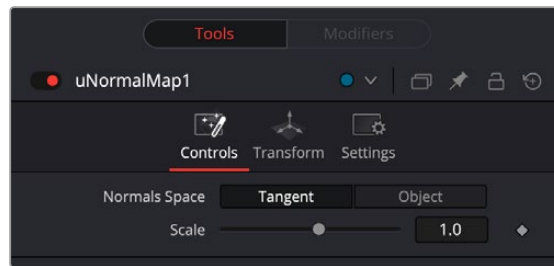


The USD Normal Map

Introduction

The `uNormalMap` tool provides several controls to modify a “normals” texture before it's passed on to the `uShader`. This allows you to manipulate the texture in various ways and ensure that it is optimized for your specific needs. Once any necessary adjustments have been made, the modified image is then passed on to the `uShader`.

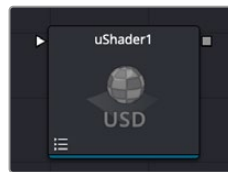
Usage



The uNormal Map controls

To use uNormalMap, place it after a uTexture node and before a uShader node. This node only works between the loaded texture and uShader.

uShader [uSd]



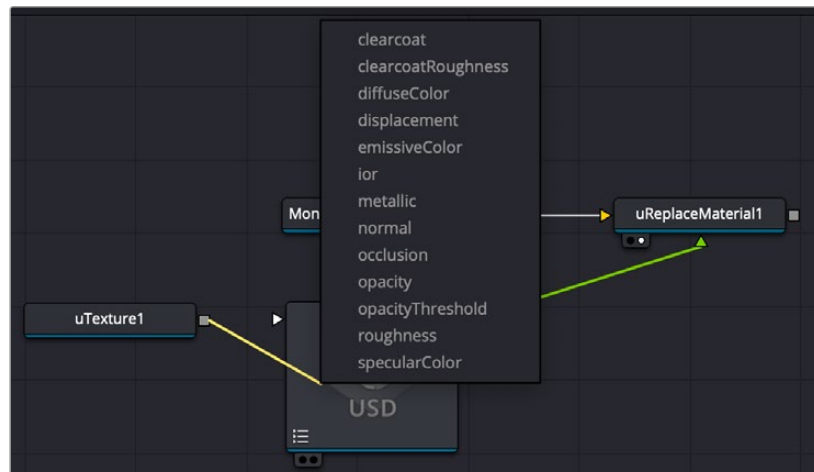
The USD Shader

uShader Node Introduction

The uShader node is used to create materials and looks using either the USD or MaterialX shader tools. Used in conjunction with uReplaceMaterial, this tool gives an easy and intuitive way to iterate material changes to a USD source.

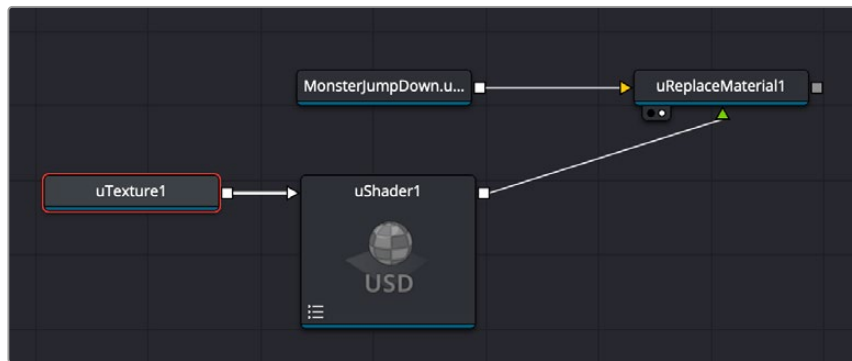
Inputs

The uShader node has variable inputs that allow you to import different assets into the tool, such as a texture, by using a uTexture node. When connecting the tool, you can choose from a drop list of all possible connections. Connecting a texture to any input will disable its corresponding slider and expose new Channel, Scale, and Bias controls to let you choose which channel the property uses, with their strength and base values.



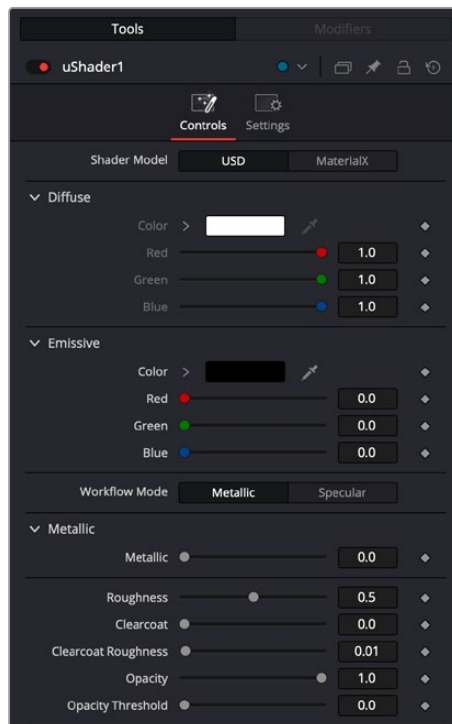
Attaching a uTexture node to the uShader node presents a drop list, showing all possible inputs to connect to.

Basic Node Setup



The uShader is accepting an input from the uTexture node, modifying that texture, then passing it to the uReplace Material node. uReplace Material is selecting specific materials from the MonsterJumpDown USD scene to apply the newly modified texture to.

Inspector



The uShader controls in USD mode

Shader Model

Here you can choose what toolset to work with, either USD or MaterialX tools.

Shader Model: USD

Diffuse Mode

Diffuse describes the base surface characteristics without any additional effects like reflections or specular highlights.

- **Color:** Provides controls to modify the shape's diffuse color.

Emissive

Emissive adds a global lighting effect to a shape, creating a layer of color over the existing texture. It can be adjusted to different levels of intensity and can be used to simulate the emission of light.

Workflow Mode

Lets you set the mode between Metallic and Specular.

- **Metallic:** When enabled, this control shows the Metallic slider. The Metallic slider adds a reflective quality to objects, creating the appearance of metal. It enhances the reflective properties of a shape.
- **Specular:** When enabled, this shows the Specular Color controls. Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears.

Roughness

Affects the amount of light reflection and scattering on a surface, giving it a smoother or more textured appearance, depending on the setting. It is a key element in creating realistic reflections.

Clearcoat

Affects the appearance of a surface by adding a glossy layer on top of it, mimicking the effect of a protective coating. It is commonly used in creating materials like car paint or polished metal.

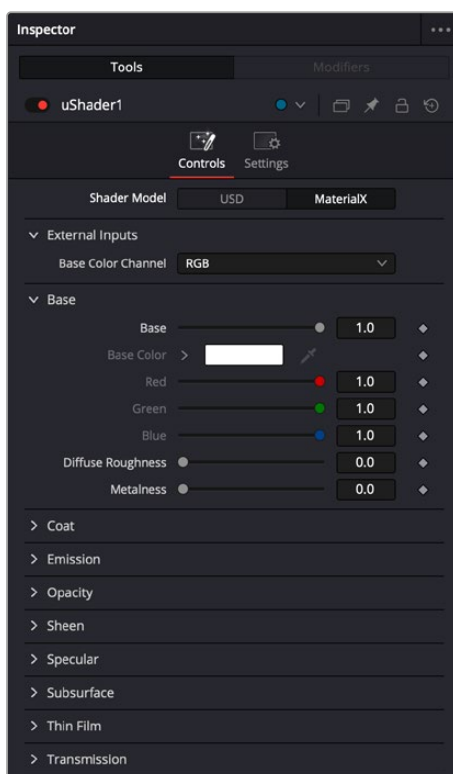
Clearcoat Roughness

Adjusts the level of imperfection or smoothness of the glossy layer added on top of a surface. This can affect the realism and reflection of materials like car paint or polished metal.

Opacity

Reduces the material and object opacity, impacting the color and Alpha values of the shape.

Shader Model: MaterialX

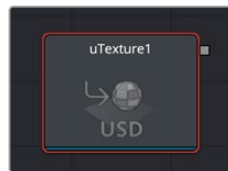


The uShader controls in MaterialX mode

Controls

MaterialX is an open standard maintained by the Academy Software Foundation designed for creating material and look development content that is platform independent. MaterialX is under constant development, and you can find the latest specifications at the MaterialX website: materialx.org

uTexture [uTx]

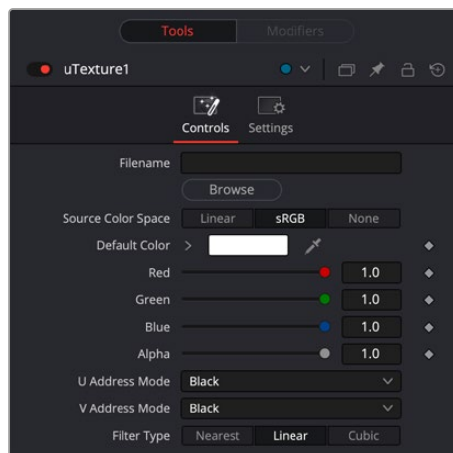


The USD Texture

Introduction

To create a material/surface in the uShader, you can import a texture file using this tool. The uTexture sources image files and uses them to build the material/surface in the uShader.

Usage

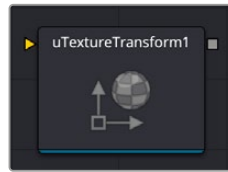


The uTexture controls

After adding the node to the Node Editor, use the Browse button to select an image file from your hard drive. Alternatively, you can click and drag an image from the Media Pool directly onto the “File Name” control in the Inspector to load an image which has already been imported into DaVinci Resolve. Use the “Source Color Space” control to choose how the image will be read by the uTexture tool. It’s important select Linear when loading a Normals image to get the best results.

U/V Address Mode controls offer a useful way to determine how the edges of the image are treated when exposed due to scaling.

uTextureTransform [uTXf]

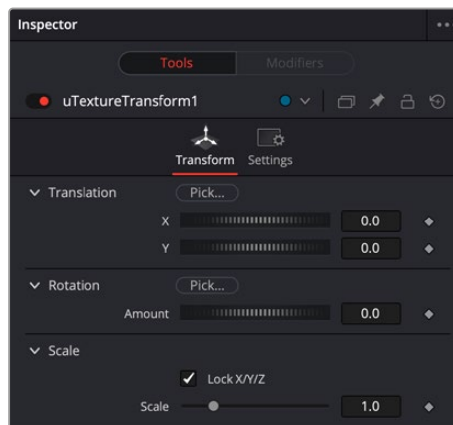


The USD Texture Transform

Introduction

uTextureTransform provides you with granular control over the scale, rotation, and position of each loaded texture file. With uTexture Transform, you can quickly and easily tweak the size, orientation, and placement of any texture file sourced from the uTexture node.

Usage



The uTexture Transform controls

To use uTextureTransform, place it after a uTexture node and before a uShader node. This node only works between the the loaded texture and uShader.

Adjust the Scale, Rotation, and Position settings to your liking.

The Scale setting allows you to increase or decrease the size of the texture file, while the Position setting lets you move the file around within your project. You can also use the Rotation setting to rotate the texture file to any angle you desire.

The Common Controls

Nodes that handle USD operations share several identical controls in the Inspector. This section describes controls that are common among USD nodes.

Transform Tab

The Transform tab can be found in the Inspector in most USD nodes. The controls are the same as the `uTransform` node, which can be applied separately.

Translation

Transform controls are used to position the USD shape in 3D space.

Rotation Order

Use these buttons to select the order used to apply the rotation along each axis of the object. For example, XYZ would apply the rotation to the X-axis first, followed by the Y-axis, and then the Z-axis.

Rotation

Use these controls to rotate the shape around its pivot point. If the `Use Target` checkbox is selected, then the rotation is relative to the position of the target; otherwise, the global axis is used.

Pivot

A pivot point is the point around which an object rotates. Normally, an object rotates around its own center, which is considered to be a pivot of 0,0,0. These controls can be used to offset the pivot from the center.

Scale

If the `Lock X/Y/Z` checkbox is checked, a single scale slider is shown. This adjusts the overall size of the object. If the `Lock` checkbox is unchecked, individual X, Y, and Z sliders are displayed to allow scaling in any dimension.

Settings Tab

The Settings tab in the Inspector can be found on every tool in the USD category. The controls are consistent and work the same way for each tool.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.